A Conversation with John von Neumann

Interviewer: Professor von Neumann, you made foundational contributions to quantum mechanics, game theory, and the modern computer. Is there a unifying thread?

Von Neumann: The thread is logic and structure. Whether you are analyzing the bluffing in poker, the probabilistic nature of a quantum state, or the architecture of a calculating machine, you are dealing with formalized systems of rules.

Interviewer: Your architecture—the von Neumann architecture—is still the basis for almost all computers today. Programs and data stored in the same memory.

Von Neumann: It was a practical necessity. If a machine is to be truly universal, as Turing described, it must be able to modify its own instructions. The distinction between data and program is merely a matter of perspective.

Interviewer: You also worked on the Manhattan Project. Some view your contributions to game theory as a cold, calculating approach to the Cold War.

Von Neumann: Game theory is an abstraction of conflict and cooperation. The Cold War was a dangerous game, yes. Mutually Assured Destruction is a terrifying concept, but mathematically, it represents a stable equilibrium. My goal was always to understand the rational basis of decisions, even in the most irrational of human endeavors.

Interviewer: What do you think of artificial intelligence today?

Von Neumann: We built the first machines to calculate ballistic trajectories. Now, they are simulating human thought. The singularity we once hypothesized—the point where technological growth becomes uncontrollable and irreversible—seems to be approaching. It is both fascinating and perilous.